

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B. Tech Mid Semester Exam

Semester: 6th SEM
 Subject Code: 303105353
 Subject Name: Machine Learning

Date: (17/02/2026)
 Time: (1hr: 30min)
 Total Marks: 40

Sr. No.		
Q.1	(A) One-line Questions: (i) What are the primary differences between the Linear and Logistic regression models? Ans: Linear Regression is used to predict continuous values (e.g., salary, price). Logistic Regression is used to predict categorical outcomes (mainly binary classes like 0/1).. (ii) What is a confusion matrix? Ans: A confusion matrix is a table used to evaluate the performance of a classification model by showing the actual vs. predicted classifications. (iii) What is the purpose of a training dataset and test dataset? Ans: The training dataset is used to train the model, while the test dataset evaluates its performance on unseen data. (iv) What is overfitting and underfitting? Ans: Overfitting occurs when a model learns the training data too well, while underfitting occurs when it fails to capture the underlying patterns in the data. (v) What is a missing value? Ans: A missing value is a data entry that is not recorded or unavailable for a particular variable in a dataset.	05
	(B) Fill in the blanks: - In linear regression, the goal is to fit a line that minimizes the "Error" between predicted and actual values, often using the method of "least squares". - The process of replacing missing values in a dataset is part of "Data Preprocessing", which ensures the data is suitable for analysis. - The "Z-score" method is a common statistical approach to detect outliers based on their distance from the mean in terms of standard deviations. - In machine learning, "hypothesis" spaces represent the set of all hypotheses consistent with the training data. - Normalization is a data preprocessing technique where the values of features are scaled to fit within a specified range, often between "0" and "1".	05
Q.2	Attempt any four (Short Questions):	12
	(1) What are the main learning paradigms in machine learning? Ans: The three main learning paradigms in machine learning are: Supervised Learning: The model is trained on labelled data to make predictions based on input-output pairs. Unsupervised Learning: The model finds patterns or structures in unlabelled data. Reinforcement Learning: The model learns through interaction with an environment,	

	receiving feedback in the form of rewards or penalties.	
	(2) Define the concept of version spaces in hypothesis evaluation. Ans: Version spaces in hypothesis evaluation refer to the set of all hypotheses that are consistent with the given training data, representing all possible solutions that fit the data.	
	(3) Evaluate the effectiveness of hyperparameter tuning in improving model accuracy. Ans: Hyperparameter tuning improves model accuracy by optimizing model parameters (such as learning rate, regularization, and tree depth) to find the best configuration for the given data, enhancing performance and reducing overfitting or underfitting	
	(4) Explain the Z- Score analysis to identify outliers? Ans: Z-Score analysis identifies outliers by calculating how far each data point is from the mean, measured in standard deviations. Data points with Z-scores greater than a chosen threshold (typically $Z > 2$ or 3) are considered outliers. The formula for Z-score is: $Z = (X - \mu) / \sigma$ where X is the data point, μ is the mean, and σ is the standard deviation.	
	(5) Describe the role of “Probability” in machine learning? Ans: Probability in machine learning model predictions quantifies the likelihood of different outcomes, allowing models to make predictions with confidence levels. It helps in tasks like classification (e.g., predicting the probability of a class) and regression (e.g., estimating the probability distribution of outputs).	
Q.3	Attempt any two questions	08
	(1) Analyze the differences between multi-class and multi-label classification, providing examples of scenarios where each is applicable. Ans: Multi-class classification and multi-label classification differ in how they handle the assignment of labels to instances: 1. Multi-class Classification: ○ Definition: Involves assigning each instance to one and only one class from multiple possible classes. ○ Example: Classifying images of animals as "cat," "dog," or "rabbit." Each image belongs to exactly one category. ○ Applicable Scenario: Handwritten digit recognition (0-9), where each image is assigned a single digit. 2. Multi-label Classification: ○ Definition: Each instance can be assigned multiple labels simultaneously. ○ Example: Categorizing a movie with labels "Action," "Adventure," and "Comedy" since a single movie can belong to multiple genres. ○ Applicable Scenario: Email categorization, where an email can be tagged as "Important," "Work," and "Urgent" at the same time.	

(2) What is PAC learning, and how does it help in evaluating the efficiency of a learning algorithm?

Ans: PAC Learning (Probably Approximately Correct Learning) is a framework in machine learning that helps evaluate the efficiency of a learning algorithm by defining conditions under which the algorithm can learn a function that approximates the target function within a specified error margin and with high probability.

Key Concepts:

1. **Probably:** The algorithm will likely produce a good model with high probability, often denoted as $1 - \delta$ (where δ is the probability of failure).
2. **Approximately:** The learned model's error will be within a specified threshold (ϵ) of the true error of the target function.
3. **Correct:** The algorithm's prediction will be accurate enough, meaning its performance will be close to the optimal or true solution.

How PAC Learning Helps in Evaluating Efficiency:

- **Generalization:** PAC learning focuses on the ability of a model to generalize well from the training data to unseen data, ensuring the model does not overfit.
- **Sample Complexity:** It provides bounds on the number of training examples needed to achieve the desired error and confidence level, thus evaluating the efficiency of the algorithm in terms of data usage.
- **Algorithmic Performance:** By determining whether an algorithm can learn within the specified error margin with high probability, PAC learning helps assess its efficiency in terms of both computational resources and the quality of learning.

(3) Given the following training data:

Point	X	Y	Class
A	1	2	Red
B	2	3	Red
C	3	3	Blue
D	6	5	Blue

A new point **P (2, 2)** is given.

Classify P using **K = 3** and **Euclidean distance**.

Ans-

	Step 1: Euclidean Distance Formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ Step 2: Distance Calculation • $d(P, A) = \sqrt{(2 - 1)^2 + (2 - 2)^2} = 1$ • $d(P, B) = \sqrt{(2 - 2)^2 + (2 - 3)^2} = 1$ • $d(P, C) = \sqrt{(2 - 3)^2 + (2 - 3)^2} = \sqrt{2} \approx 1.41$ • $d(P, D) = \sqrt{(2 - 6)^2 + (2 - 5)^2} = 5$ Step 3: Select K = 3 Nearest Neighbors A (Red), B (Red), C (Blue) Step 4: Majority Voting • Red = 2 • Blue = 1  Predicted Class = Red 	
Q.4	(A) A dataset contains values: [10,12,15,18,100]. (i) Calculate the mean and standard deviation. (ii) Compute the Z-Score for each value. (i) Calculate the mean and standard deviation: Mean (μ): The mean is calculated by summing all the values and dividing by the total number of values. $\mu = \frac{10 + 12 + 15 + 18 + 100}{5} = \frac{155}{5} = 31$ Standard deviation (σ): The standard deviation measures how spread out the values are. It is computed as the square root of the average of the squared differences from the mean. - First, calculate the squared differences from the mean: $(10 - 31)^2 = (-21)^2 = 441$ $(12 - 31)^2 = (-19)^2 = 361$ $(15 - 31)^2 = (-16)^2 = 256$ $(18 - 31)^2 = (-13)^2 = 169$ $(100 - 31)^2 = (69)^2 = 4761$ - Then, calculate the variance (the average of the squared differences): $\text{Variance} = \frac{441 + 361 + 256 + 169 + 4761}{5} = \frac{4988}{5} = 997.6$ - Finally, calculate the standard deviation: $\sigma = \sqrt{997.6} \approx 31.6$	05

(ii) Compute the Z-Score for each value:

The Z-score is calculated as:

$$Z = \frac{X - \mu}{\sigma}$$

Where X is each data point, μ is the mean, and σ is the standard deviation.

1. For $X = 10$:

$$Z = \frac{10 - 31}{31.6} = \frac{-21}{31.6} \approx -0.664$$

2. For $X = 12$:

$$Z = \frac{12 - 31}{31.6} = \frac{-19}{31.6} \approx -0.601$$

3. For $X = 15$:

$$Z = \frac{15 - 31}{31.6} = \frac{-16}{31.6} \approx -0.506$$

4. For $X = 18$:

$$Z = \frac{18 - 31}{31.6} = \frac{-13}{31.6} \approx -0.411$$

5. For $X = 100$:

$$Z = \frac{100 - 31}{31.6} = \frac{69}{31.6} \approx 2.18$$

(B) What do you mean by Data Visualization? Explain any four types of result Visualization Techniques.

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Ans: Data Visualization refers to the graphical representation of information and data. It uses visual elements like charts, graphs, maps, and plots to help users understand trends, outliers, and patterns in data. By transforming complex datasets into visual formats, data visualization simplifies the process of data interpretation and decision-making.

Four Common Types of Result Visualization Techniques:

1. Bar Chart:

A bar chart uses rectangular bars to represent data values. The length or height of each bar is proportional to the data it represents.

Example: Comparing the sales figures of five products over a specific year.

2. Line Chart:

A line chart connects data points using a continuous line. It is often used to display data trends over time.

Example: Tracking a company's revenue growth over the last 10 years.

3. Scatter Plot:

A scatter plot uses dots to represent data points on a two-dimensional axis. It shows the relationship or correlation between two variables.

	Example: Analyzing the relationship between advertising spend and sales growth. 4. Heatmap: A heatmap uses color coding to represent data values in a matrix. Darker or lighter shades indicate higher or lower values, respectively. Example: Displaying customer purchase activity across various days and times.	
	OR	
	(B) Describe the steps involved in preprocessing a dataset, including handling missing values, normalization, and adapting the data to fit a chosen algorithm. Ans: Preprocessing a dataset is essential for ensuring the data is clean, consistent, and suitable for the chosen machine learning algorithm. Below are the key steps, including handling missing values, normalization, and adapting the data to fit a specific algorithm. 1. <u>Understand and Explore the Dataset:</u> • Inspect the Data: ○ View dataset structure, data types, and missing values. ○ Identify categorical, numerical, and text data. • Summary Statistics: ○ Use metrics like mean, median, and standard deviation. ○ Visualize distributions using histograms or boxplots. 2. <u>Handle Missing Values:</u> Missing data can lead to biases or errors during modeling. Common approaches include: • Identify Missing Values: ○ Use tools like <code>isnull()</code> in Python or visualizations like heatmaps to detect missing values. • Handling Strategies: ○ Remove: Drop rows or columns with excessive missing data. ○ Impute: ▪ Numerical data: Replace with mean, median, or a predictive value. ▪ Categorical data: Replace with the mode or a placeholder (e.g., "Unknown"). ▪ Advanced techniques: Use regression or k-Nearest Neighbors (KNN) imputation.	05

3. Normalize or Scale the Data:

Scaling ensures numerical features are on a similar scale, preventing dominance by larger values in some algorithms.

- **Normalization:**

- Rescales data to a range (e.g., [0, 1]).
- Formula: $X_{\text{normalized}} = \frac{X_{\text{max}} - X_{\text{min}}}{X - X_{\text{min}}}$

- **Standardization:**

- Centers data around the mean and scales it to unit variance.

Formula: $X_{\text{standardized}} = \frac{(X - \mu)}{\sigma}$

4. Encode Categorical Variables:

Algorithms generally require numerical input, so categorical variables must be transformed:

- **Label Encoding:**

- Assign integers to categories (e.g., "Red" = 0, "Blue" = 1).

- **One-Hot Encoding:**

- Create binary columns for each category (e.g., "Red" = [1, 0, 0], "Blue" = [0, 1, 0]).

- **Target Encoding:**

- Replace categories with the mean of the target variable for each category.

5. Handle Outliers:

Outliers can skew model results:

- **Detection:**

- Use methods like the Interquartile Range (IQR) or Z-scores.
- Visualize with boxplots or scatterplots.

- **Handling:**

- Remove or cap extreme values.
- Apply transformations like log scaling to reduce impact.

6. Feature Engineering:

- **Feature Creation:**

- Generate new features from existing ones (e.g., House Age = Current Year - Year

Built).

- **Feature Selection:**

- Remove irrelevant or redundant features using correlation matrices, feature importance, or dimensionality reduction techniques like PCA.

7. Split the Dataset:

Divide the data for training, validation, and testing:

- **Train-Test Split:**

- Typically, 70-80% for training and 20-30% for testing.

- **Cross-Validation:**

- Use k-fold cross-validation for robust evaluations.

8. Adapt the Dataset to the Chosen Algorithm:

Different algorithms have unique requirements:

- **Linear Models:**

- Ensure linear relationships between features and the target.

- **Tree-Based Models:**

- Handle missing values and categorical data; scaling is often unnecessary.

- **Deep Learning:**

- Normalize data and ensure a sufficient amount of training data.

- **High-Dimensional Data:**

- Apply dimensionality reduction techniques (e.g., PCA, t-SNE).